

BOM requirements

What the schematic has to buy, and the simulation that says so.

Every row here exists because a block in `sim/blocks/` failed without it. The rule this file follows is the same one the netlists follow: **a figure nobody has read in a datasheet does not become a design number**. So these are requirements stated as specifiable, purchasable properties — tempco bands, ESR bands, on-resistance ceilings — not manufacturer part numbers. Naming a part is a separate step, and it needs a retrieved datasheet.

Where a part number *does* appear below, it is because it was checked.

Input protection

TVS — transient suppressor, battery input

SMCJ43CA. Bidirectional, 43 V standoff, 1500 W package.

Both halves of that were forced by simulation, not chosen:

- **43 V standoff** is the smallest standard bin above the 40 V clamped load dump. Below it the part conducts during a *normal* ISO 7637-2 pulse 5b event rather than a fault — which is what `load_dump.cir` was failing on, not "the TVS is too small."
- **1500 W (SMC) package, not 600 W (SMB)**. At the SMB size, pulse 2a clamps at 88.6 V against the buck's 100 V absolute maximum — 11% margin. The SMC part has roughly half the dynamic resistance at the same standoff and clamps at 73.3 V, 1.37×.

Checked by `load_dump.cir` (0.000 A, 0.000 J through the TVS) and `transient_clamp.cir`.

NEG-CLAMP — negative active clamp, buck VIN

N-channel MOSFET, $V_{ds} \geq 80\text{ V}$ (100 V class), $R_{ds(on)} \leq 10\text{ m}\Omega$ at $V_{gs} = 4.5\text{ V}$, source to ground, drain to VIN, driven by a comparator sensing VIN against a -0.1 V threshold. **In parallel with** the 100 V / 20 A clamp Schottky, which stays as the instant passive backstop.

CORRECTED 22 September 2026 — both parts were under-rated for the rail they sit on. This tag originally read $V_{ds} \geq 40\text{ V}$ and the backstop Schottky was 60 V. Both have a terminal on `VBAT_PROT`, and both were sized for the *negative* pulses they exist to catch — but pulse 2a drives the **same node** to +73.3 V, where a 40 V FET is 0.55× rated and a 60 V Schottky 0.82×. The 40 V FET had zero margin on the 40 V load dump

alone. `transient_clamp.cir` already carried the rule "every part on VBAT_PROT must clear 73.3 V" — as a principle, not a list, so nothing compared these two against it. It is now a table in `run_sim.py`: every part with a terminal on this rail, its rating, and the margin, and a part below the line fails.

The requirement is really an *effective clamp impedance*, and it is worth stating as one: the LM5164's VIN absolute maximum is -0.3 V , pulse 1 delivers 150 V through $10\text{ }\Omega$, so

$$R_{\text{clamp}} \leq 0.3\text{ V} / 15\text{ A} = 20\text{ m}\Omega, \text{ junction drop included}$$

No junction device meets that, and the reason is structural rather than a sizing miss. Decomposing the Schottky's own drop at 15 A:

term	value
junction, $N \cdot V_t \cdot \ln(I/I_s)$	0.367 V
bulk, $I \cdot R_s$	0.180 V
total	0.547 V

Only the bulk term falls when parts are paralleled; the junction term is logarithmic in current and barely moves. Four Schottkys in parallel still leave 0.375 V. A FET in its ohmic region has no junction offset at all — 15 A through $5\text{ m}\Omega$ is 75 mV.

Checked by `negative_pulses.cir`: buck VIN reaches -0.068 V on pulse 1 ($4.4\times$ inside the rating) and -0.027 V on pulse 3a ($11\times$), against -0.528 V and -0.393 V with the Schottky alone.

Closed 22 September 2026 — and closing it found a latch the block had been reporting as a pass.

The control is TLV3201-Q1 (50 ns max, push-pull, no phase inversion beyond the rails) into UCC27517A-Q1 (4 A, 23 ns max at 12 V, output held low in UVLO). The comparator's 52 mA would take about half a microsecond to charge the clamp FET's gate; the driver takes nanoseconds.

The clamp is a low-side ideal diode, and it has to let go. The model engaged it near $V_{\text{IN}} = +0.08\text{ V}$ and released it only above $+0.12\text{ V}$. At the end of every negative pulse the source returns to $+13.5\text{ V}$ and drives current back through the still-on clamp — 1.35 A through $5\text{ m}\Omega$ is 6.7 mV, never reaching the release threshold — so **the clamp latched on and shorted the source for the rest of the run.** `run_sim.py` checked only the clamp branch's *minimum* and reported it RESOLVED. A recovery check added the same day failed against it on both pulse 1 and pulse 3a. With a real battery instead of the test generator's $10\text{ }\Omega$, that is a dead short after the first negative pulse.

The thresholds are now **engage below -100 mV , release above -6.8 mV** , from the drawn network and checked as arithmetic against it:

sense node	10 k Ω from VIN, 499 k Ω from 5V_MAIN: $0.9804 \cdot \text{VIN} + 98.2\text{ mV}$, so -100 mV arrives as 0 V and no negative reference is needed
reference	10 k Ω to ground, 536 k Ω from the output: 0 V low, 91.6 mV high
release, worst case	-6.8 mV nominal, -1.8 mV with the comparator's full 5 mV offset — still below zero, so it cannot hold on against reverse current (523 k Ω , the first value, sat at $+0.5\text{ mV}$)

The window before it engages is now a number, and it is inside the rating. With the $\sim 100\text{ ns}$ delay ($50 + 23\text{ ns}$ max plus gate charge) and a 10 ns timestep cap, VIN crosses -100 mV , keeps falling $\sim 90\text{ ns}$, and bottoms at **-0.177 V** — inside the LM5164's -0.3 V by **1.70 $\times$** . The Schottky backstop barely conducts. The previous $-0.068\text{ V} / 4.4\times$ was optimistic by 2.6 $\times$: at the default step, ngspice committed the hysteretic switch on a rejected trial step and hid the dip.

Supplies are the power-up argument. The comparator runs from 5V_MAIN, which is dead whenever VIN is too low to run the buck — so it cannot hold the clamp on and short an incoming battery. The driver runs from 12V_GATE, held up $\sim 400\text{ ms}$ by the boost reservoir, so the clamp keeps its gate through a whole pulse.

Clamp Schottky

100 V / 20 A class — it was 60 V; see the correction above. It carries what the TVS used to: **13.58 A peak, 14.5 mJ** on pulse 1, measured at its own ammeter in `negative_pulses.cir` rather than inferred from 150 V / 10 Ω .

A 100 V Schottky has a higher forward drop than the 60 V part `negative_pulses.cir` models (`SCHN` , BV = 60 V). That drop only matters for the comparator's propagation delay — the window before the active clamp engages — and is re-measured when the comparator is chosen.

Reverse-battery FET

P-channel, $|\text{Vds}| \geq 80\text{ V}$, $\text{Rds(on)} \leq 40\text{ m}\Omega$ at $\text{Vgs} = -4.5\text{ V}$, drain to the battery side, source to the load, with a **passive gate network**: 47 k Ω gate-to-ground and a 12 V zener gate-to-source.

CORRECTED 22 September 2026 — the sheet had drawn an N-FET, and `reverse_battery.cir` has always modelled a P-FET. Its header says "a P-FET held on by a controller that watches the polarity", and its switch model closes on positive polarity. Found while choosing the controller the N-FET would have needed.

No controller IC checked can sit on this node. It swings from +73.3 V (pulse 2a, after the TVS) to -47.8 V (pulse 1, the bidirectional TVS's negative clamp — bidirectional precisely so a reversed battery does not forward-bias it). That is a 121 V span:

Controller	Rated	Fails by
LM74700-Q1	ANODE -65 V to +65 V	8.3 V on the positive side
LM5050-1-Q1	IN -0.3 V to +100 V	47.5 V on the negative side — and IN draws 320 μ A, so a series resistor to protect it corrupts its 20 mV sense

A P-FET needs no controller: $V_{gs} = -V(\text{load})$, so it is on whenever the load side is positive and off when it is pulled to zero. That is the block's switch model, with no IC ratings on a 121 V swing to violate.

What it gives up, recorded rather than hidden. An ideal-diode controller blocks reverse *current*; this blocks reverse *polarity*. On a dropout that *shorts* the input, bulk capacitance discharges backward through the FET. The spec carries no hold-up requirement, and the negative clamp is already sized for this FET staying on through a whole pulse — its turn-off via 47 k Ω and C_{iss} takes tens of microseconds.

The 40 m Ω ceiling gives 0.2 V at 5 A against the block's 0.3 V requirement. The model's 8 m Ω is its header's own "mid-range for a 100 V P-channel" and is optimistic; the ceiling is what `run_sim.py` checks.

Power

PDN-BULK — 3V3_MCU bulk capacitor

Polymer aluminium electrolytic or polymer tantalum, 150 μ F, specified by an ESR band of 20–50 m Ω across -40...+125 $^{\circ}$ C — not by a 25 $^{\circ}$ C typical figure.

Three things about that are load-bearing:

- **The part class changes, not the value.** A wet aluminium or tantalum electrolyte loses conductivity as it freezes, roughly 4 $\times$ ESR at -40 $^{\circ}$ C, which puts the PDN peak at 197 m Ω against a 100 m Ω target. A solid polymer electrolyte has no such mechanism.
- **The cold number is what gets specified.** A supplier's 25 $^{\circ}$ C ESR does not answer this block's question, and asking for it is how the original corner got in.
- **A band, not a ceiling.** Bulk ESR is also what damps the regulator-against-bulk resonance. At 20 m Ω the peak climbs back to 92 m Ω , 8 m Ω under target. "ESR $\leq$ 50 m Ω " would be satisfied by a 5 m Ω part that fails this block.

The 100 mΩ target is derived — 50 mV allowed ripple over a 0.5 A transient step — not a round convention. Checked by `mcu_pdn.cir`.

SENSOR-PTC — 5 V sensor rail, one per sensor group

Resettable PTC, hold current ≥ 200 mA, R25 = 3.0 Ω nominal.

Sized against the ceiling rather than guessed. The polyfuse has to be what clears a shorted harness, not the 5 V regulator's own current limit, so fault current must stay under 3.0 A:

$$\begin{aligned} I < 3.0 \text{ A} &\Rightarrow R_{src} + R_{ptc} > 5/3 = 1.667 \text{ } \Omega \\ \text{corner} &= R_{25} \times 0.8 \text{ (cold)} \times 0.7 \text{ (-30\% spread)} = 0.56 \times R_{25} \\ &\Rightarrow R_{25} > 2.78 \text{ } \Omega \end{aligned}$$

At the same stacked $-40^\circ\text{C} + -30\%$ corner the specified part draws 2.88 A, 0.12 A under the ceiling. The larger series resistance was re-checked against the rest of the block rather than assumed harmless: healthy-group voltage holds at 4.69 V against a 4.5 V floor, and nominal fault current stays inside the 0.3–3.0 A window the polyfuse needs to trip on.

Checked by `sensor_rail.cir`.

Drivers

MU-ISENSE — metering unit current sense, ECU pin 88 return

50 mΩ shunt on the low side, into either a current-sense amplifier feeding an ADC channel or a gate driver with an integrated current-regulation comparator.

The amplifier is not optional: 50 mΩ at the 0.675 A setpoint is 33.75 mV, too small to take straight to the S32K148's ADC at useful resolution.

This one is not a part-selection fix at all — it is an architecture change, and it is the only failing corner in the suite that no component choice could have answered. The coil is copper, copper moves 0.39%/°C, and over $-40\dots+125^\circ\text{C}$ its resistance goes $7.39 \rightarrow 13.82 \text{ } \Omega$. At a fixed 50% duty the mean current follows it: **0.862 A cold against 0.462 A hot**, +27% and -32% around the setpoint. There is no better resistor to buy.

A hysteretic current regulator holds 0.679 / 0.679 / 0.683 A at $-40 / 25 / +125^\circ\text{C}$. **Closed loop only works while the regulator has duty left** — at the hot corner the coil can draw at most $13.5/13.82 = 0.977$ A at 100% duty, so the setpoint sits at 69% of the ceiling. That bound is checked, because a regulator out of headroom degrades silently back to open-loop behaviour rather than reporting anything.

Checked by `metering_unit_pwm.cir`.

EGR-DRIVER — EGR H-bridge

A bridge driver with **internal DIR/PWM decode** — one logic input pair that cannot express both legs of a diagonal conducting at once.

Two logic lines wired straight to two diagonal switch pairs short the bridge: with both inputs asserted it draws **270 A**. Under a decoded interface the same input combination draws **4.35 A**, the motor's own stall current. The fault state stops existing rather than being avoided by firmware convention.

Current capability: headroom over a 2.6–6.4 A stall.

Drawn 22 September 2026 — discrete, two AUIRS2184STR. No integrated bridge clears the rail: DRV8873-Q1, the candidate, has a 40 V VM absolute maximum against 73.3 V, and the class caps at 40–45 V. So four **100 V** N-FETs ($\leq 20\text{ m}\Omega$, headroom over the 6.4 A stall), and the decode this tag requires comes from the **driver's structure** rather than a logic gate: AUIRS2184S has **one input per leg** — IN high puts HO on, IN low puts LO on — with cross-conduction prevention and $\sim 400\text{ ns}$ deadtime. Each leg's high and low are exclusive by construction, so the 270 A state, both inputs high, becomes a **high-side brake**.

EGR_EN	IN1 IN2	bridge
high	1 0 / 0 1	forward / reverse
high	1 1 / 0 0	high-side / low-side brake
low	any	COAST

Coast is asserted by copper. Both drivers' $\overline{\text{SD}}$ share EGR_SD, pulled down $10\text{ k}\Omega$, so with PTB10 Hi-Z at reset both legs are off and the return spring closes the valve. A kill FET on the same node gives coast on GATE_KILL without the MCU. One $20\text{ m}\Omega$ shunt in the shared low-side return feeds an INA181A1-Q1: 2.56 V at a 6.4 A stall.

Pinout and logic are from the IR2184 datasheet (same die) and the AUIRS2181(4)S family table; **the automotive part's own datasheet was not retrieved** — confirm against it at sourcing.

Checked by `egr_hbridge.cir`, which keeps the naive wiring as pinned evidence of why the requirement exists.

GATE-PULLDOWN — fail-safe gate pulldowns, six driver gates

$470\text{ }\Omega$, gate to source, on every driver gate.

The largest standard value that still holds a gate below $V_{gs(th)} = 1.0\text{ V}$ against Miller coupling at $C_{rss} = 500\text{ pF}$. The commonly-reached-for $10\text{ k}\Omega$ fails even at the *low* end of the memo's own C_{rss} envelope — 1.155 V at 50 pF already exceeds the threshold.

Cost, stated because it is real: 21.3 mA per gate held at 10 V, 1.28 W across all six.

Checked by `supervisor.cir`.

Signal

`CAN-TERM` — CAN split termination, Rah / Ral

60.0 Ω **thin-film**, initial tolerance ±0.1% or better, tempco ±50 ppm/°C or better.

Ordinary standard-grade thick-film runs 100–200 ppm/°C. At 200 ppm/°C over the full –40/+125 °C span, both resistors shifting together, the differential impedance reaches 123.95 Ω against J1939's own 120 Ω — and 120 Ω here is what the standard specifies, not a round convention, so missing it is a real finding.

The contrast is a part class, with no meaningful cost or availability penalty at this quantity. With both mechanisms stacked the adverse way ($60 \times 1.00825 \times 1.001 = 60.5555 \text{ } \Omega$) the specified part lands at 121.10 Ω, inside the ±2 Ω window with 0.90 Ω to spare. Checked at the **worst case**, not the nominal — so passing means the requirement is sufficient, not merely that it helps.

Checked by `can_termination.cir`.

`VR-CLAMP` — crank input clamp diodes, four off

Reverse leakage ≤1 μA at 125 °C. Forward drop is explicitly *not* a requirement.

This inverts the rule every earlier clamp on this board was chosen by. The negative-clamp Schottky's 0.547 V at 15 A decomposes into a junction term and a bulk term, and that decomposition is what ruled out paralleling junction devices for the power path. These four diodes conduct only on a fault, into a 4.7 kΩ series resistor that is already doing the limiting — what matters is what they do when they are **off**.

A reverse-biased diode's leakage flows out of the 4.489 kΩ Thévenin at the comparator input and appears there as an offset voltage, against a hysteresis window of ±198 mV:

Class	Leakage at 125 °C	Offset	Against the window
BAT54S-class Schottky	~200 μA (≈2 μA at 25 °C, ~100× hotter)	0.90 V	4.5× the whole window — latches
BAV199-class low-leakage silicon	<1 μA (5 nA max at 25 °C)	<4.5 mV	2% of the window

The two diodes on a leg are reverse-biased symmetrically by `VR_BIAS`, and the legs are matched, so leakages partly cancel and what survives is partly common-mode. That is not a thing to defend a speed input on, so the requirement is stated instead.

`vr_conditioner.cir` does **not** model this — its comparator is an ideal switch with no input network, so no clamp, no leakage and no offset exist in it. Found while drawing `hw/speed_inputs.kicad_sch`.

The same tag covers the **differential TVS across the pair**: bidirectional, standoff ≥ 48 V. The standoff has to clear the *signal*, not just the rail — VR amplitude is proportional to speed, about 20 V peak at 1500 rpm, so a 24 V part would clip a runaway engine's own crank signal at the one moment the ECU most needs to keep counting teeth. 48 V clears 3600 rpm.

VR-COMP — crank zero-cross comparator

Single 3.3 V supply, **push-pull** rail-to-rail output, propagation delay ≤ 1 μ s. Input common-mode range: **mid-rail only**.

Chosen 22 September 2026: TLV3201AQDCKRQ1 — the same part as the negative clamp's comparator, so one part number serves both. 2.7–5.5 V, push-pull, input range (VEE) – 0.2 V to (VCC) + 0.2 V, **50 ns max** against the 1 μ s requirement, 5 mV offset against a ± 198 mV window.

Three of those four requirements are unusually easy, deliberately:

- The output swing *is* the hysteresis window ($3.3 \text{ V} \times 4489 / (4489 + 33000) = 0.395 \text{ V}$), so an open-drain part would make the window depend on its pull-up value and its V_{OL} .
- Biasing the floating VR coil to `VR_BIAS` means the inputs never leave mid-rail, so **no common-mode range including ground is needed** — the usual hard constraint on a zero-cross detector, avoided by biasing the sensor rather than level-shifting it.
- 1 μ s is 0.009 crank degrees at 1500 rpm on a 60-tooth wheel.
- Input offset voltage and its tempco are non-issues: 10 mV is 5% of the window.

Checked by `vr_conditioner.cir`, which establishes the topology (a fixed 5 V threshold produces edges at 1500 rpm and **none** at cranking speed) and the ± 0.2 V window the E24 network above lands on.

BOOST-BLEED — injector boost reservoir bleed

22 k Ω , 1 W, across the 47 μ F / 100 V reservoir.

The charger regulates from **one side only**: `boost_converter.cir`'s `Bchg` adds 50 mA while the rail is below setpoint and adds nothing above it. Nothing else in the circuit can remove charge, so any excursion above 100 V is permanent. `injector_turnoff.cir` measured the mechanism that produces one — a hold-current cutoff, whose energy came from the battery through the high side's diode-OR and so has no boost-side draw to net against — and its RESULT NOTE asked for this resistor by name.

Sizing is not a duty-cycle argument; it is "one event is gone before the next cylinder fires":

event charge	10 A decaying to 0 over 20 μ s = 100 μ C
on 47 μ F	2.14 V of rail rise
above setpoint	charger is off , so the whole 100 V / 22 k Ω = 4.55 mA discharges it
time to clear	21.4 ms , against 26.67 ms between cylinders at 1500 rpm
cost	455 mW , burned continuously whenever the rail is up

Checked by `boost_converter.cir` arrangement C, which runs two identical reservoirs — one bare, one bled — through three such events at the real cylinder spacing. Bare, the rail stacks to 106.4 V and stays there.

A comparator-gated shunt costs nothing at idle and is the alternative. It is more parts, and it is a part nobody has chosen.

Over a complete injection event there is no pumping at all — the peak phase draws 342 μ C from the reservoir and end-of-injection recirculation returns roughly 100 μ C of it. This resistor is not there for normal running. It is there because one-sided regulation has no answer for an excursion from *any* cause.

INJ-RECIRC — injector recirculation diodes, five off

150 V, 20 A, **ultrafast**. Three on the low sides (one per cylinder, into the boost reservoir) and two on the high sides (the battery diode-OR).

Research memo 10 names the class. Not a Zener active clamp across the FET — Nexperia AN50003 rates repetitive active clamp reliability- "Questionable" for gate-oxide wear-out. Not a bare freewheel to battery, which is too slow: the same problem the boost rail exists to fix on the opening side.

Sized by the pulse, not the average. `injector_turnoff.cir` measures 28.7 W instantaneous during turn-off against roughly 3.6 W averaged over all nine events per second — a factor of eight. The average would never have sized this part.

Chosen parts

These are the entries that have moved from "a class" to "a part number", each against a datasheet that was actually retrieved and read. They are listed separately from the requirements above because they are answers, not constraints.

Function	Part	Why this one
Buck pre-regulator	LM5164DDAR	100 V input, 1 A, 6–100 V range. See below — it replaces a 60 V part that can no longer survive this input.
3V3 regulator	TLV76733QWDRBRQ1	1% over load <i>and</i> temperature. The supervisor makes that load-bearing.
Supervisor	TPS3850G33DRCT	±4% window variant, so its undervoltage trip clears the MCU's own LVD by 143 mV rather than 44 mV.
Injector boost controller	TPS40210QDGQRQ1	Chosen on its 200 ns maximum off-time — the one spec that makes 94% duty reachable at the 6 V cranking corner. See below.
Gate drivers, all eight gates	AUIRS2181STR ×5	No cross-conduction interlock — the injector's high and low switches are in series with the coil and must both be on. See below.
Difference amplifiers	INA592IDR ×5	Native $G = \frac{1}{2}$, 88 dB CMRR min against a 60 dB floor the discrete network could not reach. Not AEC-Q100 — see below.
Barometric sensor	MPXHZ6115A6T1	Spec §7 deviation 2, on-PCB behind a vented port. AEC-Q100 not stated in its datasheet — see below.
EGR bridge drivers	AUIRS2184STR ×2	One input per leg: each leg's high and low exclusive by construction. See EGR-DRIVER .
Negative-clamp control	TLV3201AQDCKRQ1 + UCC27517AQDBVRQ1	A low-side ideal diode: engage below −100 mV, release above −6.8 mV. See NEG-CLAMP .
Current-sense amplifiers	INA181A1QDBVRQ1 ×3	VCM reaches −0.2 V, so it sits across a ground-referenced shunt; supplied from 3V3_MCU so its output is ADC-safe by construction.
Kill FETs and reset inverter	2N7002BK ×10	60 V, ±20 V Vgs, Vth 1.1–2.1 V, 2 Ω max at Vgs 5 V against the 5 Ω supervisor.cir assumes. AEC-Q101. Needs GATE-KILL-CLAMP .
GATE_KILL clamp	BZX84-C12-Q	11.4–12.7 V at 5 mA, +10 mV/K max. AEC-Q101. The plain BZX84 series is no longer automotive-qualified; the -Q suffix is.
Relay FETs	PMV280ENEA ×2	100 V, ±20 V Vgs, 432 mΩ max at Vgs 4.5 V. AEC-Q101. Driven through a buffer — see RELAY-GATE .
Relay gate buffers	CAHCT1G125QDBVRQ1 ×2	SN74AHCT1G125-Q1 on 5V_MAIN: VIH 2 V reads a 3.3 V GPIO, VOH 4.4 V min drives the FET. AEC-Q100.
Reverse-battery FET	SQD50P08-25L	−80 V P-channel, ±20 V Vgs, 31 mΩ max at −4.5 V against a 40 mΩ ceiling. AEC-Q101, TO-252.
100 V N-channel	IPD90N10S4-06 ×8	Negative clamp, metering, four EGR, two injector battery switches (60 V class needed; one part number instead of two). 6.7 mΩ max at 10 V, ±20 V Vgs, AEC-Q101, TO-252. The logic-level sibling -S4L-06 was rejected for its ±16 V Vgs: every one of these gates is driven from 12V_GATE.
Injector 150 V FETs	SQM85N15-19 ×5	Two boost-side high sides, three low sides. 19 mΩ max at 10 V — the model's 20 mΩ. ±20 V Vgs, AEC-Q101, TO-263. Qg 120 nC max sets the bootstrap droop: 126 mV.

Function	Part	Why this one
Boost switch	DMN15H310SK3	150 V, Vth 3 V max, 350 mΩ max at Vgs 4 V — specified below the 5.7 V it gets while cranking. Qg 8.7 nC typ. AEC-Q101, TO-252. Cranking duty with its drop, hot: 94.8% against 97.0% available.

GATE-KILL-CLAMP — 12 V zener on GATE_KILL

GATE_KILL is pulled up by 47 kΩ to VBAT_PROT, not 3V3_MCU, and for a good reason: a pullup on the rail being supervised has no supply during the brownout it exists to catch. But nothing on the net draws DC: it is nine FET gates and a reset inverter's drain. So with the inverter off, the net follows the battery rail, and that rail reaches 73.3 V on pulse 2a. That is 3.7× the ±20 V Vgs rating of any small-signal FET.

This was not caught while the kill path was drawn. It surfaced on 22 Sep 2026, when the first kill FET was being chosen and its Vgs rating had to be checked against what drives its gate.

A BZX84-C12-Q from GATE_KILL to ground holds the net at 12.7 V max, or 13.7 V at 125 °C. Its current at the pulse peak is 1.3 mA, and the 47k dissipates 78 mW for the pulse's duration. Below 12 V, which includes cranking, the zener is out of circuit and GATE_KILL is the battery, which is what the pullup is there for. The numbers are checked in `run_sim` `transient_clamp`.

RELAY-GATE — the relay FETs' gates at 5 V, through a buffer

The relay sheet drove each FET's gate straight from a 3.3 V GPIO, and its field note said "fully enhanced at 3.0 V". No part backed that claim. The 100 V automotive small-signal FETs checked were PMV280ENEA, DMN10H220LQ and BSS123N. Every one of them specifies Rds(on) at Vgs = 4.5 V and no lower, and their Vth runs up to 2.5–2.8 V. That leaves 0.5–0.8 V of overdrive, less when cold, and the datasheets guarantee nothing there. The 100 V class is not optional here, because the drain is on VBAT_PROT through the coil.

So an SN74AHCT1G125-Q1 on 5V_MAIN sits between the pin and the gate. Its TTL inputs read 3.3 V logic (VIH 2 V at VCC 4.5–5.5 V), and its output is 4.4 V minimum. OE is tied low. A 100 k pulldown on the input keeps the relay off while the GPIO is Hi-Z from reset. The existing 10 k pulldown on the gate covers the case where 5V_MAIN is absent. At 84 mA of coil current, the PMV280ENEA drops 36 mV.

INA592IDR — the difference amplifiers, and a stronger argument

TI SBOS914F, October 2018, revised April 2021. Retrieved and text-extracted. Five, one per differential channel on `sensors_analog`.

Gain	native $G = \frac{1}{2}$ (±IN inputs, SENSE tied to OUT)
CMRR	88 dB minimum

Gain error	±0.03 % max
Supply	4.5–36 V single; runs from 5V_MAIN
Output	within 220 mV of each rail
Input	common mode to $3 \cdot (V+) - 2 \cdot \text{REF} = 15 \text{ V}$ at $G = \frac{1}{2}$; only absolute maximum is 10 mA, and the ~12 kΩ internal network holds a 40 V harness short to about 3.3 mA

Why a difference amplifier and not an op-amp. The requirement was $\geq 60 \text{ dB CMRR}$, the floor `sensor_differential.cir` tried. The drawn design — an op-amp with a discrete 26k/16k network — could not reach it: 0.1 % discrete resistors give about 48 dB. The network has to be matched on-die. $G = \frac{1}{2}$ also puts a 0.5–4.5 V sensor at 0.25–2.25 V, straight into the ADC.

And the argument got stronger with a real number. The block had modelled 80 dB, marked INFERRED, at gain 16/26. With the chosen part's **minimum** 88 dB at $G = \frac{1}{2}$ — and both paths' errors referred to the sensor, since they no longer share a gain — differential now beats single-ended by 394× on a 1 Ω corroded return, up from 90×.

A 1 kΩ now sits between each amplifier and its ADC tail: the amplifier runs from 5 V and can rail at 4.78 V on a harness fault, where the 3.3 V zener at the pin would otherwise take the amplifier's full short-circuit current. With 1 kΩ it takes 1.5 mA.

Not AEC-Q100, and nothing better was found. INA2132 was the dual alternative, but its " $G = \frac{1}{2}$ " (datasheet Figure 12) is a single-ended attenuator; its matched network only gives a *difference* $G = 1$, which pushes a 4.5 V sensor past its 4 V swing on 5 V. No qualified difference amplifier with a differential $G = \frac{1}{2}$ turned up.

MPXHZ6115A6T1 — the onboard barometric sensor

NXP (Freescale) MPXA6115A series data sheet, Rev. 7.3, April 2015, retrieved and text-extracted. Spec §7 deviation 2 adds it on purpose: "on-PCB behind a vented port; costs nothing in harness terms and improves fuelling correction."

Range	15–115 kPa absolute
Transfer	$V_{OUT} = V_S \times (0.009 \cdot P - 0.095)$, P in kPa
Supply	4.75–5.25 V, 10 mA max
Accuracy	±1.5 % of span, 0–85 °C; compensated –40...125 °C
Package	super small outline, case 98ARH99066A — a footprint KiCad ships

Supplied from 5V_MAIN, which `SENSOR_5V_MON` already measures — so the reading is ratiometric-correctable with no new channel. Through the same 10k/16k divider every 5 V channel on the sheet uses: 2.51 V at

sea level (101.3 kPa), 1.92 V at 2000 m. No zener at the MCU pin: every other channel on that sheet arrives from the harness; this never leaves the board.

AEC-Q100 is not stated in the retrieved datasheet. Infineon's KP236 is listed automotive-qualified with a 40–115 kPa range and would be the choice, but its datasheet did not resolve by any route this session — Mouser timed out twice, Infineon's own host redirected. It is a **footprint change**, not a drop-in, so the choice is recorded rather than guessed: confirm qualification at sourcing, or retrieve the KP236 sheet and swap.

The **vented port** is a mechanical requirement the schematic cannot carry.

INA181A1QDBVRQ1 — one part, one gain, three channels

TI SLYS018F, April 2018, revised October 2024. Retrieved and text-extracted.

Common mode	−0.2 V to 26 V — covers a ground-referenced shunt
Gain	20 V/V (A1), ±1 % max over temperature
Offset	±150 µV max at VCM = 0 V, 1 µV/°C max drift
Bandwidth	350 kHz (A1)
Supply	2.7–5.5 V; output swings to VS − 0.02 V
CMRR	84 dB min

Channel	Shunt	Full scale	Where it sits
Injector bank A → PTC15	5 mΩ	1.80 V at 18 A peak, 1.00 V at 10 A hold	ADC0
Injector bank B → PTD19	5 mΩ	same	ADC1 — simultaneous with bank A
Metering → PTC16	50 mΩ	675 mV at the 0.675 A setpoint, 844 counts	ADC0

The gain is chosen for the injector. 1.80 V at the peak leaves headroom to 3.28 V, so a fault current up to 32.8 A is measured rather than clipped — which is what an overcurrent check needs. Offset is worth 30 mA of injector current (0.17 % of peak) and 3 mA of metering current (0.44 % of setpoint). Bandwidth resolves the 38 µs ramp to peak with about 0.45 µs of lag.

Supplied from 3V3_MCU, which makes the output ADC-safe by construction. It cannot swing above VS − 0.02 V, so these three channels need no 3.3 V zener at the MCU pin — unlike every channel on sensors_analog , which arrives from the harness.

This closes `MU-ISENSE` 's open half. That tag specified the 50 mΩ shunt and said "the amplifier below is not optional"; this is the amplifier. `IN-` goes to GND, and the tag's layout requirement stands: a Kelvin tap at the shunt's own ground pad, not a via into the pour.

`AUIRS2181STR` — chosen because it has *no* interlock

Infineon (International Rectifier) `AUIRS2181(4)S`, datasheet dated 10 January 2014, still hosted by Infineon. Retrieved and text-extracted this session. Its first listed typical application is "Piezo / common rail Injection." Confirm lifecycle status at sourcing — the datasheet is old even though the part is still listed.

Offset	VB 625 V absolute maximum; VS operational −5 V to +600 V
Supply	VCC 10–20 V; UVLO+ 8.0 / 8.9 / 9.8 V max
Logic	VIH 2.5 V min, VIL 0.8 V max — direct from a 3.3 V MCU pin
Output	1.9 A / 2.3 A typ, 1.4 A / 1.8 A min short-circuit at 15 V
Delay	160 / 200 ns typ, 270 / 330 ns max
Interlock	None on 2181/21814 (the 2183/2184 have it)
Package	SOIC-8, AEC-Q100

Three properties chose it, each against a part that failed on it:

Property	This part	Why it matters	Failed it
No cross-conduction interlock	none	In a half bridge, HO and LO together is shoot-through and every driver prevents it. Here the "high" and "low" switches are in series with the coil and must both be on for it to conduct at all	UCC27282-Q1, UCC27712-Q1 — could not fire an injector
Offset rating	625 V abs max, 5.4× margin	Bank node ~103.8 V worst case, VB 12 V above it = 115.9 V	UCC27211A-Q1 — HB 120 V abs max, 4.1 V of headroom
Negative VS	operational to −5 V, clears by 3.7 V	<code>D_fw</code> drops the bank node 1.0–1.3 V below ground on every hold off-time	UCC27211A-Q1 — HS DC min −1 V, violated in normal operation

The first is the one that is not a rating, and the one nobody searching a parametric table would filter for: the familiar reflex is that a half-bridge driver *without* interlock is the less safe choice. In this circuit interlock is not a safety feature, it is a failure to function.

Five in use. Four on `injector` — each package pairs one high side with one low side, by package and not by circuit; UB2's low side is spare. One on `metering_egr`, low side only, so the board carries one gate-driver part number instead of two.

The kill path moves to `HIN`. The part has no shutdown pin, and a ground-referenced FET cannot short a gate floating 100 V up. `HIN` is ground-referenced, so `supervisor.cir` 's kill FET goes there behind a 1

kΩ from the MCU. It is slower than the low-side gate clamp — 330 ns maximum driver turn-off against 23 ns — and that is acceptable because it is not the path the fail-safe argument rests on: the low sides are in **series** with every coil, and killing any one stops that cylinder.

Checked by `boost_converter` 's rating checks in `run_sim.py` .

GATE-RAIL — 12V_GATE, a discrete follower off the boost rail

The board had no rail between 5 V and 40 V, and the drivers need 10–20 V. **Its source is the boost rail, and the reason is cranking:** a regulator from `VBAT_PROT` cannot make 12 V from a 6–9 V cranking battery, and a gate rail that collapses during cranking means no injection exactly when the engine needs it. `BOOST_100V` is the one rail already designed to hold up from 6 V.

Not the LM5164 already on the board — its VIN absolute maximum is 100 V (TI datasheet §5.1), and this rail sits at ~102 V.

Part	Value	Why
Bias	$2 \times 47\text{ k}\Omega$	87 V split so neither resistor carries all of it; 0.87 mA
Reference	13 V zener	$13\text{ V} - V_{be} - \text{limit drop} = 12.07\text{ V}$ at 5 mA
Pass	NPN, $V_{ceo} \geq 160\text{ V}$, DPAK, AEC-Q101	0.44 W at 5 mA; a shorted output puts the whole rail across it
Limit	small-signal NPN + $56\text{ }\Omega$	$0.65\text{ V} / 56\text{ }\Omega = 11.6\text{ mA}$, holding a short to 1.16 W
Output	10 μF 25 V	Holds through each driver's gate-charge pulse

Load is about 3 mA (five drivers' quiescent current plus gate charge), designed for 5 mA. **12V_GATE clears the drivers' UVLO+ ceiling by 2.27 V.** Standing load on the boost is now 9.65 mA — bleed, gate rail and divider — of its 50 mA, and the peak-phase draw still recovers in 8.48 ms against 26.67 ms between cylinders.

Its failure runs the safe way. If the boost stops, 12V_GATE collapses, every driver drops into UVLO and holds its outputs low, and every gate on the board turns off.

TPS40210QDGQRQ1 — chosen on an off-time, not on an input range

TI SLVS861F, August 2008, revised June 2020. Retrieved and text-extracted this session.

Input	4.5 V–52 V, absolute maximum 52 V (Table 6.1)
Reference	700 mV, 686–714 mV over $-40/+125\text{ }^{\circ}\text{C}$ (Table 6.5)
Off-time floor	tOFF(min) = 170 ns typ, 200 ns max (Table 6.6)
UVLO	4.25 V typ, 4.5 V max turn-on (Table 6.5)
Supply current	1.5 mA typ, 2.5 mA max , not switching
Internal regulator	BP = 8 V typ (7–9 V), 0–15 mA

Gate driver	400 mA source / sink
Package	HVSSOP-10 PowerPAD (DGQ), 3 × 3 mm, AEC-Q100 grade 1

The number that chose it is the off-time, and the reasoning runs backwards from the usual. A boost from a 6 V cranking dip to 100 V needs

$$D = 1 - 6 / (100 + 0.9) = 94.05 \%$$

and a controller's minimum off-time is a floor **in time**, not in duty. At 150 kHz, 200 ns is 3 % of the period, so $D_{max} = 97.0 \%$ and the cranking corner clears by **2.95 points**. At 2.2 MHz — where most wide-input boost controllers in this class sit, and where the magnetics would be far smaller — the same 200 ns is 44 % of the period and D_{max} falls to **56 %**, below even the 86.6 % the rail needs at a *nominal* 13.5 V battery.

So "higher switching frequency, smaller inductor" is the wrong optimisation here, and the part that looks less modern is the one that works.

THE RATING THAT DOES NOT FIT — the third time on this board.

VDD's absolute maximum is 52 V; `VBAT_PROT` reaches 73.3 V for about 50 µs on ISO 7637-2 pulse 2a. Over by **1.41×**.

This is the same shape as DRV8873-Q1's 40 V `VM`, which took that part off the EGR bridge — and it is solvable here for a reason that did not apply there. A motor bridge's `VM` carries the motor current; this pin carries 2.5 mA plus gate charge. So it gets its own clamp:

BOOST-VDD-CLAMP — 47 Ω, 1 W, and a 43 V zener, 1 W

Pulse 2a, 73.3 V	zener passes $(73.3 - 43)/47 = 645 \text{ mA}$, for 50 µs = 1.39 mJ
Load dump, 40 V for 400 ms	43 V clamp does not conduct at all
Cranking, 6 V in	$2.5 \text{ mA} + 25 \text{ nC} \times 150 \text{ kHz} = 6.25 \text{ mA}$, dropping 0.29 V → $VDD = 5.71 \text{ V}$
against UVLO turn-on max 4.5 V	1.21 V of margin

43 V rather than 39 V is the whole point of the value. A 39 V clamp would sit in conduction for the load dump's entire 400 ms at roughly 0.8 W. A 43 V clamp only works during the 50 µs pulse, where 1.39 mJ is nothing.

The 47 Ω is squeezed from both ends: small enough that 6.25 mA does not drop the part below UVLO at the cranking dip, large enough that the zener sees 645 mA and not more during the pulse.

The power stage is untouched by any of this — the inductor and FET see the rail directly, and are 150 V class for it, the same class the injector switches already carry.

Checked by `boost_converter.cir` 's rating checks in `run_sim.py` — not with a switching model, which that block deliberately does not have, but as executable arithmetic on the datasheet's own numbers. Same treatment `transient_clamp.cir` gives the 73.3 V rail.

What is NOT settled by a datasheet: the loop compensation. `Rcomp` 10 k Ω and `Ccomp` 6.8 nF put the error amplifier's zero at 2.3 kHz, below the right-half-plane zero a boost puts at $(1-D)^2 \cdot R_{load} / (2\pi \cdot L) = 17.3$ kHz at 13.5 V in — and that zero **moves with duty cycle**, so the cranking corner is the hard one. These are a starting point. A current-mode boost's loop depends on the inductor's real DCR, the capacitor's real ESR and the FET's real switching behaviour; it gets measured on the bench.

LM5164DDAR — and the 60 V part it replaces

Research memo 07 §4 selected **TPS54360B-Q1**, reasoning that a 60 V part beat a 42 V LM5175-Q1 "for load-dump margin". That reasoning was correct when it was written. It is not correct now, and the change came from the input-protection work above:

- The TVS standoff had to rise to 43 V so the part stops conducting during a *normal* clamped load dump.
- A higher standoff clamps higher. Pulse 2a now clamps at 73.3 V.
- `transient_clamp.cir` carries a **passing** check that says so in as many words: *"60 V-class parts are NO LONGER viable downstream."*

A 60 V buck on this input is a part that fails the first ISO 7637-2 pulse 2a event. The LM5164 was already the part every rating check in the suite was written against — its -0.3 V / 100 V VIN limits are what `negative_pulses` and `transient_clamp` check — but nothing had ever reconciled the BOM table with it. **This is what a contradiction between two documents looks like when only one of them is executable.**

Operating range 6–100 V covers the spec's own 6–40 V requirement, with the 6 V end matching the cranking dip the input range was built around. 1 A against an estimated 400–500 mA board load.

Configuration values, all derived from the datasheet rather than picked:

Part	Value	From
<code>RON</code>	31.6 k Ω	The tON table in §5.5 is linear in RON/VIN: at 12 V, 25 k Ω gives 830 ns, so $K = 398$ ns·V/k Ω . For 400 kHz at 5 V out, $RON = tON \cdot VIN / K$ is 31.4 k Ω at both 13.5 V and 40 V — the VIN feedforward is what holds the frequency fixed. E96 value 31.6 k Ω .
FB divider	38.3 k Ω / 12.1 k Ω	VREF is 1.2 V (1.181–1.218, $\pm 1.5\%$), so 5 V needs $1 + R1/R2 = 4.167$.
EN/UVLO divider	75 k Ω / 24.9 k Ω	Enable rises at 1.5 V typ; 0.249 ratio puts turn-on at 6.02 V, the spec's cranking-dip floor. The pin is rated to 100 V, so the divider programs the threshold rather than protecting the pin.

Part	Value	From
BST cap	2.2 nF 50 V X7R	Not a range. The datasheet specifies the part: <i>"a high-quality 2.2nF 50V X7R ceramic capacitor between BST and SW"</i> .
L / Cout	33 µH / 47 µF, 10 mΩ ESR	buck_preregulator.cir, sized at the 40 V worst case rather than nominal. The ESR is part of the value — it sets the ripple.

The RON figure is worth one more line: it independently reproduces buck_preregulator.cir's own gate timings, 925 ns at 13.5 V and 312.5 ns at 40 V. The netlist got those from $D = V_{out}/V_{in}$; the datasheet gets them from RON. They agree.

TLV76733QWDRBRQ1 — accuracy as a requirement

Memo 07 §6 named TLV1117-33QDCYRQ1, explicitly *"class pricing, not fetched"*. Choosing the supervisor turned 3V3 accuracy into a hard constraint: outside 3.143–3.459 V the board resets itself. TLV767-Q1 is 1% over load and temperature — 3.267–3.333 V, about 125 mV of margin at each end. 1 A, 2.5–16 V in, AEC-Q100, VSON-8.

The fixed-output version is the one specified. Its pin 3 is SNS where the adjustable version has FB, and they are not interchangeable. SNS must not float; it is tied to OUT.

TPS3850G33DRCT

See docs/pinmap.md §1.8 and hw/gen_symbols.py. The variant letter is the decision: G sets thresholds at $\pm 4\%$ of nominal and H at $\pm 7\%$, and with the part's $\pm 0.8\%$ accuracy that is a worst-case undervoltage trip of 3.143 V against 3.044 V. The supervisor has to assert before the S32K148's own LVD (3.0 V max), so G33 clears it by 143 mV and H33 by 44 mV — close enough that the two could fire in either order, and a supervisor that might lose the race to the thing it supervises is not doing its job. The cost is the tighter rail window above.

MCU

NXP S32K148, FS32K148HAT0MLQT — 144-pin LQFP, 80 MHz, $-40\dots+125\text{ }^{\circ}\text{C}$.

Confirmed in research memo 05 against the S32K1xx data sheet Rev. 15 with page-level citations. The full reasoning is there; what matters for the BOM:

- The 144-pin package is headroom against unknown U1 (50 of the OEM ECU's 94 pins are still unresolved), not a requirement of the confirmed channel budget, which fits inside a 100-pin part.
- LQFP over MAPBGA: identical instance counts — CAN/ADC/FTM/I²C are silicon-level, not package-level — and 0.5 mm pitch is inspectable by eye or AOI, where a BGA needs X-ray or destructive inspection. A real

cost for a small team doing early assembly.

- **80 MHz H grade, not 112 MHz U**: the U speed grade is *not valid* at the 125 °C M temperature grade, per the datasheet's own ordering-table footnote. Memo 05 §6 shows the extra clock is not needed to close the timing budget, so there is no reason to trade thermal margin for headroom the design does not use. This corrects the spec §5 framing, which cites 112 MHz.
- In stock at DigiKey's India storefront at single-unit quantities. Not a lead-time part. The S32K146 fallback was checked and is *not* needed — it is the one that is out of stock, with 12–52 week lead times.

Still to check before footprint lock: package drawing 98ASS23177W (SOT486-2) has not been independently fetched. The PCB roadmap's warning stands — a hand-sourced footprint is the most likely place to introduce a silent error.